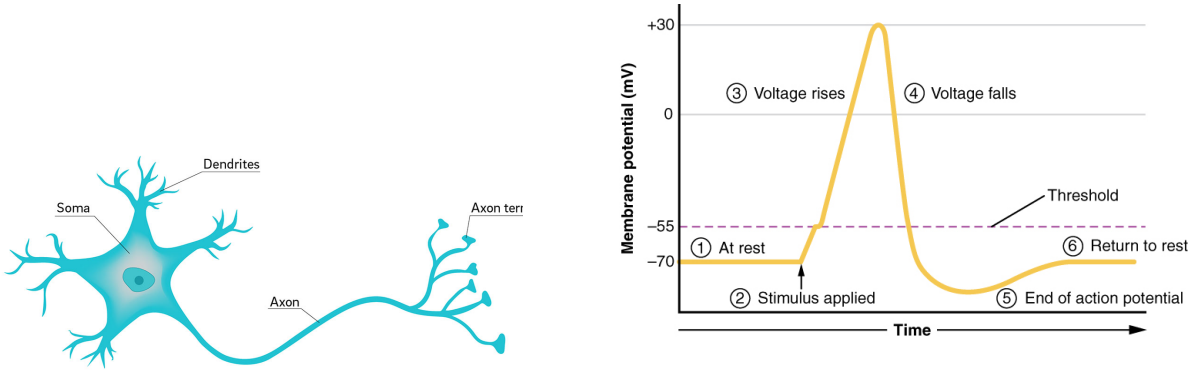
ECS4003 Advanced Computer Engineering
Laboratory & Coursework

This coursework asks you to design and implement an efficient accelerator on FPGA, including definition of the accelerator's behaviour, implementation on FPGA and optimisation of cost and real-time performance. It will use Simulink to define the accelerator behaviour, and Xilinx Model Composer (which is really just a library of special blocks for Simulink) to implement the accelerator.

Modelling Brains

Neuromorphic computer architectures attempt to model the behaviour of the human brain by constructing networks of neurons. Unlike the neurons used in artifical neural network (such as those used in deep learning), these neurons are *spiking*, and their networks known as Spiking Neural Networks (SNNs). The general structure of a neuron and its is shown in Fig. 1.



(a) Neuron Structure

(b) Behaviour

Figure 1: Models of Neurons

The *dendrites* act as inputs to the main cell body or *soma*. They transport streams of electrical current, in the form of potentials, from connected neurons. These accumulate an action potential in the soma. When the voltage reaches a critical value, the soma discharges, its internal voltage and a potential spike is output of the *axon* to neighbouring neurons.

An example model of this behaviour is defined by the equations below.

- The neuron has a single input J representing an input current, but which you can just considered a stream of input data values. It maintains local variables $v[n]$ and $u[n]$, which it combines with J and constants a, b to determine updated values $v[n+1]$ and $u[n+1]$

$$v' = v[n] + \left(0.04v[n]^2 + 5v[n] - u[n] + J[n] + 140\right) \quad (1)$$
$$u' = u[n] + a\left(bv[n] - u[n]\right) \quad (2)$$

- The neuron applies a thresholding operation so that if v' exceeds an arbitrary value (in this case 30), it 'resets' to a given value (in this case -65). When this happens, u' has a small constant d added. The values of internal variables v and u are updated according to this thresholding.

$$v[n+1] = \begin{cases} -65 & \text{if } v' > 30 \\ v' & \text{otherwise} \end{cases} \tag{3}$$

$$u[n+1] = \begin{cases} u' + d & \text{if } v' > 30 \\ u' & \text{otherwise} \end{cases} \tag{4}$$

- The output s of the neuron is boolean-valued: it takes value 1 if the reset operation has taken place, otherwise it outputs zero.

$$s[n] = \begin{cases} 1 & \text{if } v' > 30 \\ 0 & \text{otherwise} \end{cases} \tag{5}$$

Part I - Thinking, Slowly

a) Using Simulink realise (1) - (5) in the contents of the *spiking* box in the simple starter model provided. Take the following coefficient values:

- $a = 0.02$
- $b = 0.2$
- $d = 8$
- initial value of $v[0] = -65$
- initial value of $u[0] = 0$

Use **only** two-input add, subtract, product components along with delay, constant and switch components. Verify the spiking behaviour of the neuron. Save the model as *spikingTest.slx*.

b) Given the latencies for your add, subtract, product and switch components, determine the initiation interval of your spiking component.

c) Modify your model to use timed versions of the add, subtract, product and switch components as appropriate to your student number. Save the model as *timedSpikingTest.slx*.

d) State and demonstrate the latency (u.t.) and throughput (samples/u.t.) of your timed component.

e) Create a modified version, *timedSpikingTestOpt1.slx*, of c) which minimises latency.

f) Create a modified version, *timedSpikingTestOpt2.slx*, of d) which minimises the cumulative delay length in your model.

Submission
Your submission should consist of a series of Simulink models, and a recording of a presentation created using Microsoft Teams. Please note that the primary method of assessment is the presentation, and you should be careful to demonstrate that you have satisfied the assessment criteria there, including documenting models, tests and rationale and empirical measures of performance and cost.

- Each of the models named in a) - f).
- *Presentation*: A 10-15 minute (max) presentation recorded using Microsoft Teams which details:
  - Presentation of the model and evidence of successful operation for a)

  - A record of your component latencies for b), and a statement of your calculated initiation interval.
  - Presentation of the model and evidence of successful operation for c)
  - Statement and demonstration for d) (clearly stating your definition of u.t.)
  - Statement and demonstration of correct operation for e) and f), clearly stating any transformations employed and their effect.

- A handwritten derivation of the initiation interval for b).